

Mini Research #3: Network Topology and Information Spread

Research Question: How does network topology affect the spread of information in a simulated social network?

1. Introduction

Information spreads through connections between individuals. The arrangement of those connections can determine whether information remains localized, moves rapidly through the network, or reaches a large proportion of nodes only after many transmission steps. This project compares Random, Small-World, and Scale-Free synthetic networks to study these effects under controlled simulation assumptions.

2. Problem Statement

Networks with the same number of individuals can behave differently because their connections are organized differently. The experiment tests whether structural differences produce measurable changes in the speed or final extent of information spread.

3. Objective and Hypothesis

The objective is to compare information spreading across three topology models using the same main simulation assumptions. The hypothesis is that topology will affect the speed of spread even if final coverage becomes similar.

4. Methodology

Each network contains 100 nodes. The models are Erdős–Rényi Random, Watts–Strogatz Small-World, and Barabási–Albert Scale-Free. One initial node is informed. At every step, an informed node attempts transmission to uninformed neighbors with probability 0.20. Each topology is simulated for 30 steps and 30 runs using seed 42.

5. Evaluation

Final spread is the mean percentage informed after 30 steps. Time to 50% is the first step at which at least half the nodes are informed, averaged across runs.

6. Results

The Random network reached 99.93% final spread and 50% in 9.00 steps. Small-World reached 100.00% and 9.20 steps. Scale-Free reached 100.00% and 5.93 steps. The strongest difference was therefore speed rather than final coverage.

7. Discussion

Under the simulated conditions, Scale-Free was the fastest topology. A plausible structural explanation is the presence of highly connected nodes that can create many transmission opportunities. This interpretation is specific to the simulation and does not establish a universal real-world rule.

8. Limitations and Validity

The study is synthetic. The transmission probability is a modeling choice, and the generated networks have unequal edge counts: 224 Random, 300 Small-World, and 291 Scale-Free. Therefore, this is not a perfectly equal-edge controlled comparison. Results can also change with network size, transmission probability, simulation length, and graph parameters.

9. Conclusion

Under the simulated conditions, network topology affected the speed of information spread. Scale-Free reached 50% fastest at 5.93 steps, while all three networks approached complete spread by 30 steps. The result supports the hypothesis within this computational setup but should not be generalized directly to real social networks.

Main Results

Network	Final Spread (%)	Time to 50%
Random	99.93	9.00
Small-World	100.00	9.20
Scale-Free	100.00	5.93